1 A light beam of length 1.5 m is supported by a horizontal cable placed at 0.6 m from the hinge. The beam is hinged at one end supported by the wall. A weight of 300 N is hung at the other end of the beam as shown in Fig. 1.

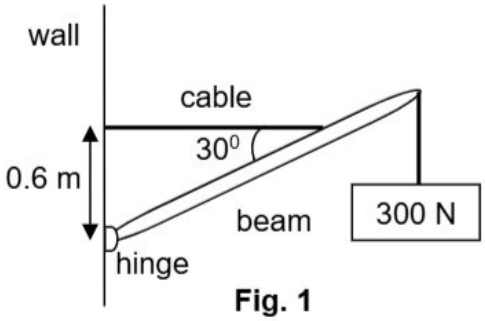


Fig. 1

(a) The three forces acting on the beam are the tension in the cable *T*, the force by the hinge on the beam *H* and the weight of 300 N.

Draw a vector diagram to represent the forces acting on the beam.

[2]

(b) Determine the magnitude of the tension in the cable *T*.

T = ... N [2]

(c) Hence determine the magnitude and direction of the force *H*.

magnitude of force *H* = ... N

direction of force *H* : ..... [3]